

Safety Filter for Toxicity Detection: Midterm Progress Report

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Abstract

We present midterm results on our toxicity detection safety filter, evaluated on 10,000 real comments from the Toxic Conversations dataset. Three baselines of increasing complexity are compared: rule-based keyword matching (F1=0.303), TF-IDF + Logistic Regression (F1=0.372, AUC=0.826), and fine-tuned DistilBERT (F1=0.616, AUC=0.959). We provide theoretical grounding for each method, detailed error analysis, hyperparameter studies, and a roadmap connecting this work to generative AI safety.

1 Introduction

Automated toxicity detection is essential for moderating user-generated content at scale. This project builds a lightweight safety filter and evaluates three baselines on real data, progressing from our initial proposal toward a robust final system.

Connection to generative AI. This work directly relates to the generative models course in several ways: (1) Safety filters serve as input/output guardrails for LLMs such as GPT and LLaMA, screening toxic prompts before generation and filtering harmful outputs before delivery. (2) Our strongest baseline, DistilBERT, is a product of generative pre-training—its contextual representations are learned via masked language modeling, a generative objective. (3) We plan to use generative models for data augmentation and adversarial robustness testing in the next stage.

2 Theoretical Background

2.1 Problem Formulation

Given text $x \in \mathcal{X}$, we predict $y \in \{0, 1\}$ (non-toxic vs. toxic) by learning $f : \mathcal{X} \rightarrow \{0, 1\}$ that minimizes expected risk $R(f) = \mathbb{E}[\ell(f(x), y)]$. With imbalanced data (8.2% toxic in our dataset), standard accuracy is misleading, so we focus on precision, recall, F1, and ROC-AUC.

2.2 Rule-Based Keyword Matching

A set $\mathcal{W}$ of bad words classifies text as toxic if any token matches: $f(x) = \mathbf{1}[\text{tokens}(x) \cap \mathcal{W} \neq \emptyset]$. This requires no training and runs in $O(n)$ per document, but cannot capture context (“kill the process” vs. “kill yourself”) or morphological variants (“stupidity” vs. “stupid”).

2.3 TF-IDF + Logistic Regression

TF-IDF maps text to sparse vectors $\mathbf{v} \in \mathbb{R}^{|V|}$ where $\text{TF-IDF}(t, d) = \text{TF}(t, d) \cdot \log(N/\text{DF}(t))$. Logistic regression models $P(y=1 \mid \mathbf{v}) = \sigma(\mathbf{w}^T \mathbf{v} + b)$ and minimizes regularized cross-entropy:

$$\mathcal{L} = -\frac{1}{N} \sum_i [y_i \log \hat{y}_i + (1-y_i) \log(1-\hat{y}_i)] + \frac{1}{2C} \|\mathbf{w}\|^2 \quad (1)$$

The convex loss guarantees a global optimum, and balanced class weights address imbalance by weighting the minority class higher. Bigram features capture limited local context (e.g., “not good”), but word order beyond pairs is lost.

2.4 DistilBERT

DistilBERT [Sanh et al., 2019] retains 97% of BERT’s performance while being 60% faster. Its core is multi-head self-attention: $\text{Attention}(Q, K, V) = \text{softmax}(QK^T/\sqrt{d_k})V$, which captures long-range dependencies and context.

The model is pre-trained via masked language modeling—a *generative* objective: $\mathcal{L}_{\text{MLM}} = -\mathbb{E}[\sum_{i \in \mathcal{M}} \log P(x_i \mid x_{\setminus \mathcal{M}})]$. This forces the model to learn rich contextual representations from large unlabeled corpora, which then transfer to downstream classification. Knowledge distillation from BERT combines supervised loss, KL-divergence on soft predictions, and cosine alignment of hidden states.

For classification, a linear head on the [CLS] representation is fine-tuned with AdamW (lr= 2×10^{-5} , warmup, weight decay).

3 Experimental Setup

3.1 Dataset

We use the **Toxic Conversations** dataset from HuggingFace (`SetFit/toxic_conversations`), containing real user comments labeled as toxic or non-toxic. We sample 10,000 comments and split 90/10 into 9,000 training and 1,000 validation samples. The toxic class is 8.2% of the data—a realistic and challenging imbalance (Figure 1).

This is a significant improvement over our Week 3 baseline, which used a synthetic dataset of 1,000 samples from hand-written templates.

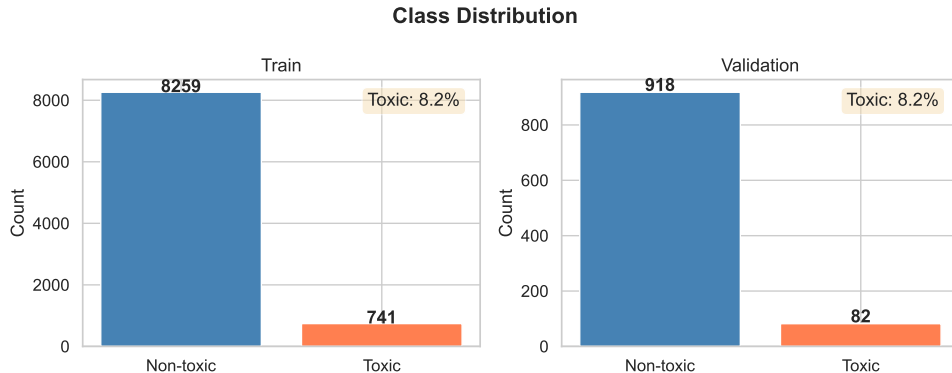


Figure 1: Class distribution: 8.2% toxic, reflecting real-world imbalance.

3.2 Implementation

All models use our `src/baselines/` codebase:

- **Rule-based** (`rule_based.py`): 36-word bad-words list, regex tokenization.

- **TF-IDF+LR** (`tfidf_lr.py`): unigrams+bigrams, 50K features, balanced weights, $C=1.0$.
- **DistilBERT** (`distilbert_clf.py`): `distilbert-base-uncased`, 3 epochs, batch 16, lr 2×10^{-5} , AdamW.

Evaluation uses `src/evaluation.py` (precision, recall, F1, FPR, FNR, ROC-AUC).

4 Results

4.1 Baseline Comparison

Table 1 shows results on the 1,000-sample validation set.

Table 1: Baseline comparison on real data (1,000 validation samples, 8.2% toxic).

Model	Prec.	Rec.	F1	FPR	FNR	AUC
Rule-based	0.349	0.268	0.303	0.045	0.732	—
TF-IDF + LR	0.392	0.354	0.372	0.049	0.646	0.826
DistilBERT	0.703	0.549	0.616	0.021	0.451	0.959

Table 2: Computational cost.

Model	Train (s)	Inference (s/1K)	Parameters
Rule-based	<0.01	0.012	0 (no learning)
TF-IDF + LR	0.83	0.040	~50K features
DistilBERT	1,761	12.35	66M

DistilBERT achieves the best F1 (0.616) and AUC (0.959), improving F1 by 66% over TF-IDF+LR. However, it requires ~30 minutes of training and is ~300× slower at inference. All models struggle with recall on this challenging real-world data, with even DistilBERT missing 45% of toxic comments.

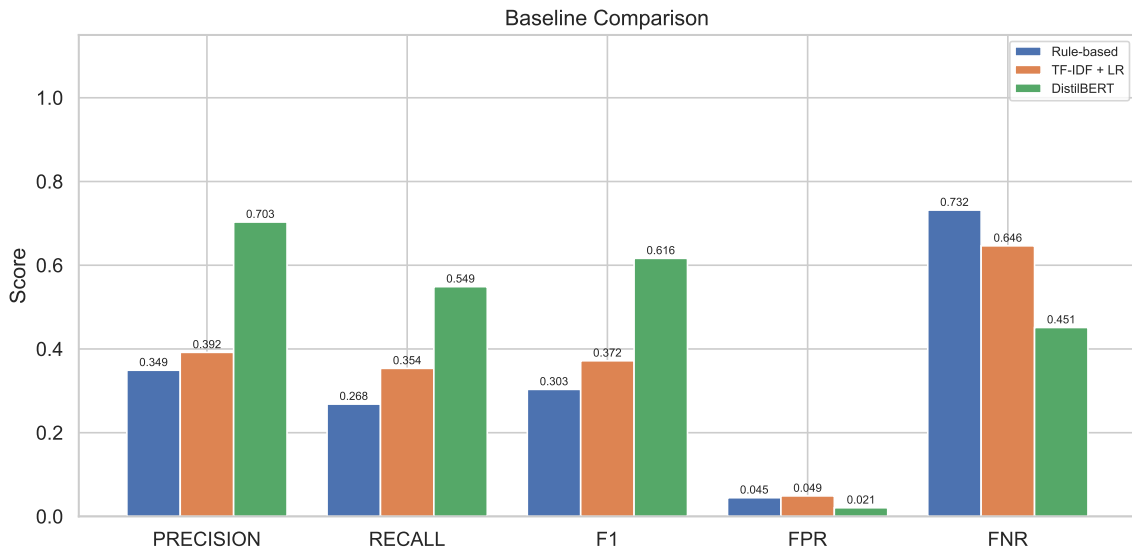


Figure 2: Key metrics across baselines. DistilBERT leads on all metrics except FPR, where rule-based and TF-IDF+LR are comparable.

4.2 Confusion Matrices

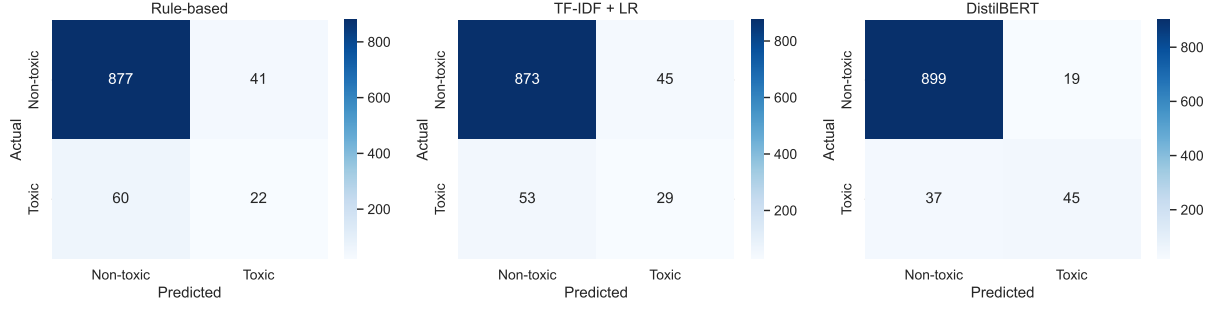


Figure 3: Confusion matrices. All models produce many false negatives due to the difficulty of detecting subtle toxicity.

4.3 ROC and Precision-Recall Curves

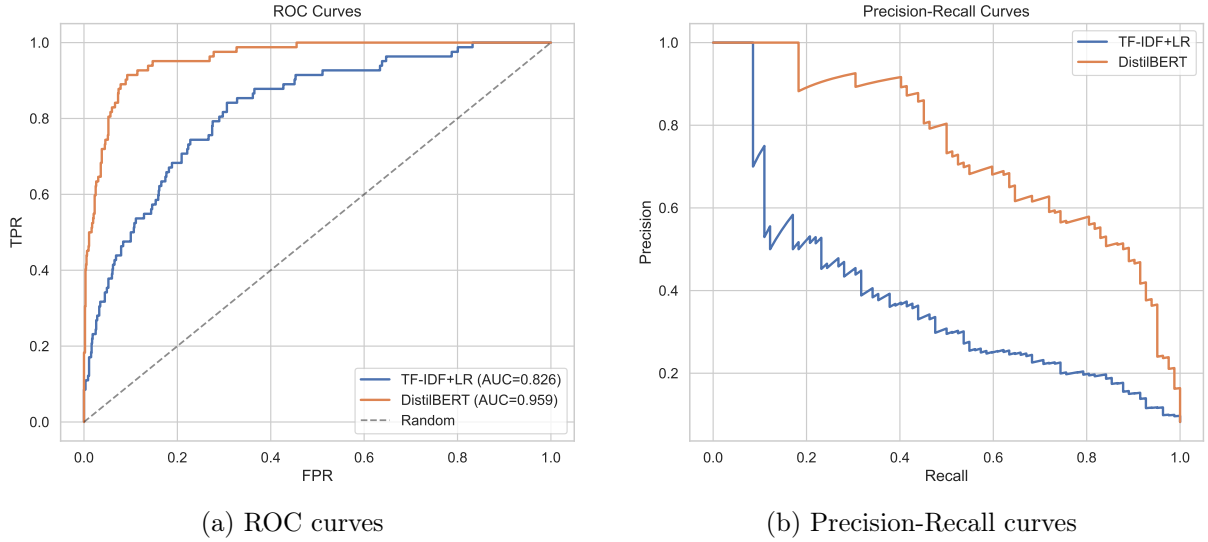


Figure 4: DistilBERT (AUC=0.959) substantially outperforms TF-IDF+LR (AUC=0.826), showing better ranking ability even when the default threshold is suboptimal.

4.4 Threshold Analysis

The default 0.5 threshold may not be optimal for imbalanced data. Figure 5 shows that lowering the threshold improves recall at the cost of precision, allowing deployment-specific tuning.

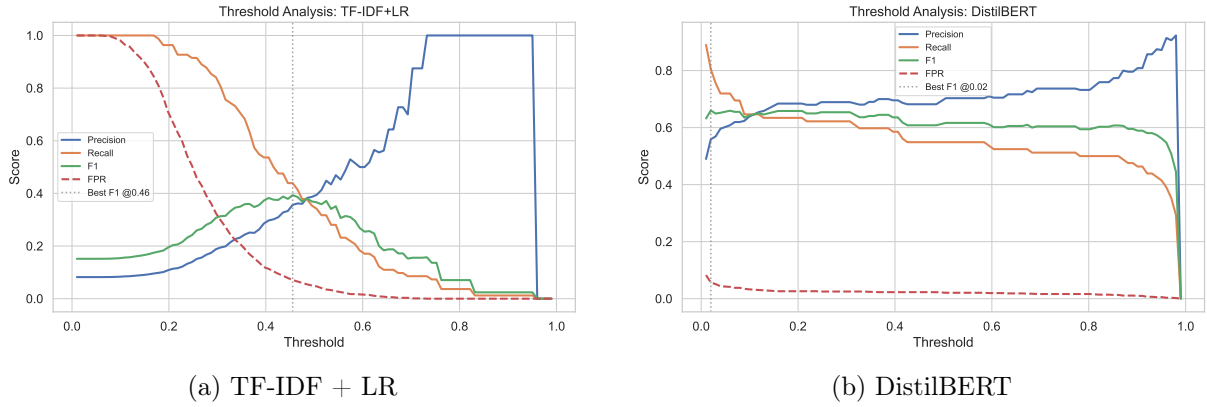


Figure 5: Threshold analysis. Lowering the threshold from 0.5 can significantly boost recall for both models.

4.5 Training Dynamics

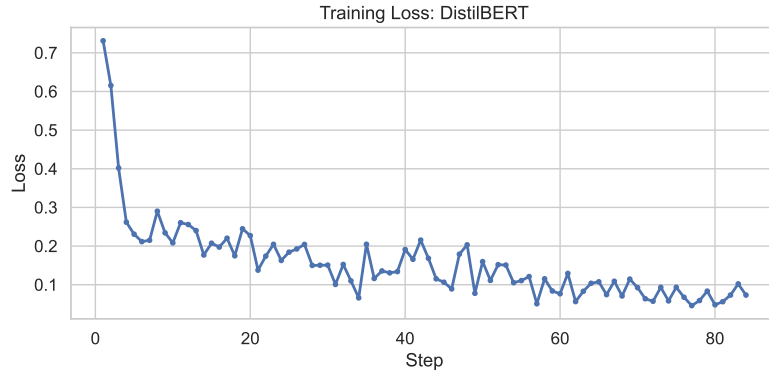


Figure 6: DistilBERT training loss over 3 epochs. Loss drops from 0.73 to ~ 0.05 , with most learning in the first epoch. The noisy loss in epochs 2–3 suggests the model is near its capacity on this data size.

5 Analysis

5.1 Error Analysis

Rule-based false negatives (73.2% of toxic missed): The keyword list cannot detect implicit toxicity. Examples: “Trump is a buffoon” (no bad word), “this guy is sort of like the schoolyard bully” (indirect). The classifier only catches explicit profanity.

Rule-based false positives (4.5% FPR): Words like “ugly” or “hate” appear in non-toxic contexts: “an ugly strain of intolerance in the nation” (quoting), “how many people die every day” (factual).

TF-IDF+LR: Slightly better recall (35.4%) by learning statistical patterns, but still misses most subtle toxicity. The model over-relies on individual toxic words without understanding context.

DistilBERT: Best at context—correctly handles many ambiguous cases. Still fails on implicit toxicity: “this guy is sort of like the schoolyard bully” (no explicit signal). False positives include borderline cases: “Diversity is a dirty word” (sarcasm), “America-hating malevolent narcissist” (political rhetoric, labeled non-toxic by annotators).

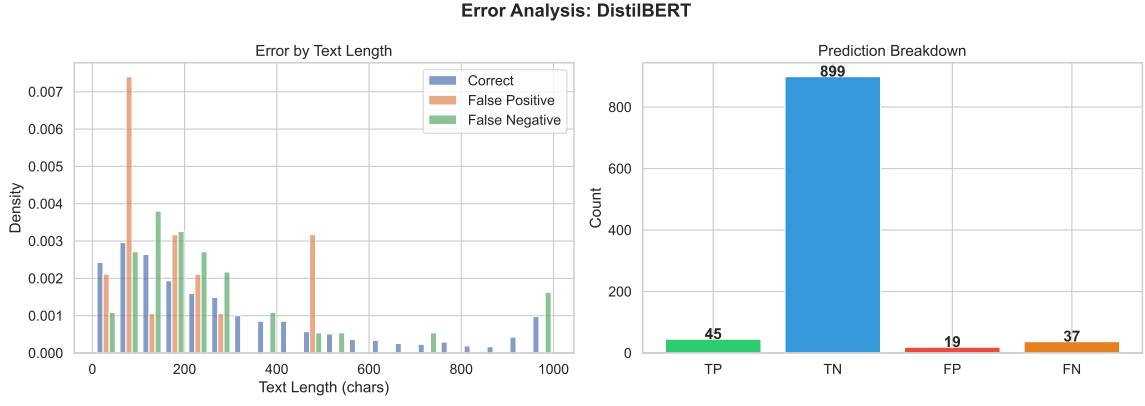


Figure 7: DistilBERT error analysis: text length distribution and prediction breakdown.

5.2 Hyperparameter Sensitivity

Figure 8 shows TF-IDF+LR performance across regularization strengths and vocabulary sizes. Unlike our synthetic baseline (where all settings gave $F1=1.0$), real data shows clear sensitivity: $C=0.5$ is optimal ($F1=0.374$), and performance degrades with strong regularization ($C=0.001$, $F1=0.325$) or weak regularization ($C=50$, $F1=0.268$). Vocabulary size also matters: 1,000 features gives $F1=0.255$, while 50,000 gives $F1=0.372$.

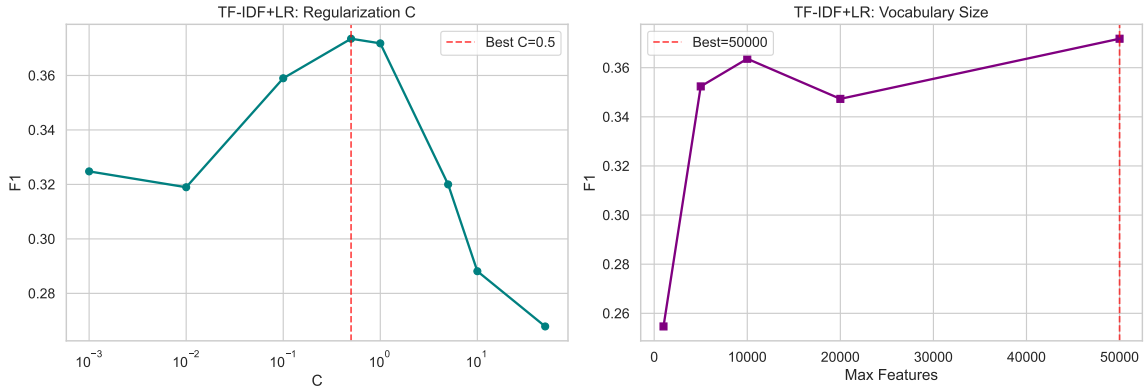


Figure 8: Hyperparameter sweep on real data shows meaningful variation, unlike the flat curves on synthetic data.

5.3 Dataset Size Ablation

Figure 9 shows TF-IDF+LR $F1$ growing from 0.023 (200 samples) to 0.372 (9,000 samples), with no plateau—indicating more training data would likely improve performance further. The rule-based classifier is constant at $F1=0.303$ regardless of data size, confirming it learns nothing from data.

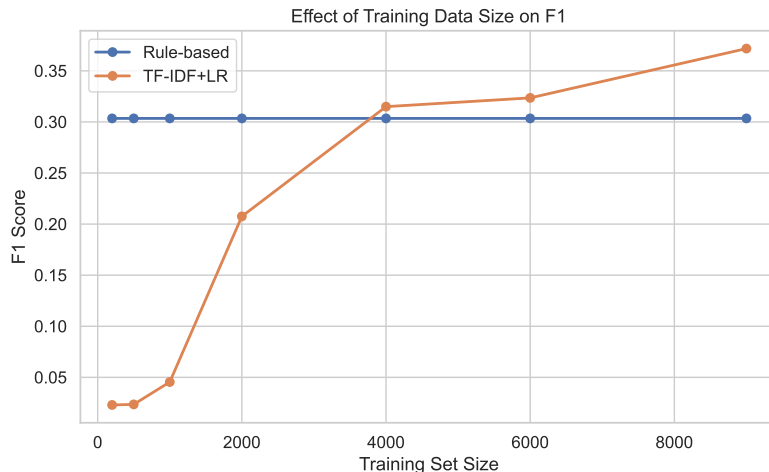


Figure 9: TF-IDF+LR benefits strongly from more data; the curve has not saturated.

5.4 Comparison with Week 3 Baseline

Table 3 contrasts our initial synthetic-data results with the current real-data evaluation.

Table 3: Progress from Week 3 (synthetic data) to Week 6 (real data).

Model	Week 3 (synthetic)		Week 6 (real)	
	F1	AUC	F1	AUC
Rule-based	0.632	—	0.303	—
TF-IDF + LR	0.556	0.834	0.372	0.826
DistilBERT	0.727	—	0.616	0.959

The drop in F1 confirms that real data is substantially harder than synthetic templates. However, DistilBERT’s AUC of 0.959 shows strong ranking ability—the model *knows* which comments are more toxic, even if the default threshold is not optimal.

6 Planned Improvements

1. **Scale to full dataset.** The learning curve (Figure 9) shows no saturation—training on the full 1.7M samples should improve TF-IDF+LR and DistilBERT substantially.
2. **Threshold optimization.** Use validation-set F1-optimal thresholds instead of the default 0.5, which is suboptimal for 8% toxic prevalence.
3. **Generative data augmentation.** Use GPT-based paraphrasing to create synthetic toxic examples, improving recall on implicit toxicity.
4. **Adversarial evaluation.** Test robustness to character substitution, leetspeak, and paraphrased toxic content.
5. **Cross-domain testing.** Train on Toxic Conversations, evaluate on Jigsaw/Civil Comments.
6. **Ensemble.** Combine DistilBERT confidence with TF-IDF+LR for speed-accuracy trade-off.
7. **Bias analysis.** Check if the classifier disproportionately flags certain dialects [Borkan et al., 2019].

7 Individual Contributions

- **Nikita Shiyanov:** DistilBERT implementation and fine-tuning, theoretical analysis, training dynamics, threshold analysis.
- **Anton Korotkov:** Rule-based and TF-IDF+LR baselines, evaluation framework, hyperparameter sweeps, ablation study, error analysis.

8 Conclusion

Moving from synthetic to real data revealed the true difficulty of toxicity detection: F1 dropped from 0.727 to 0.616 for our best model. DistilBERT’s AUC of 0.959 demonstrates that generative pre-training provides powerful representations for this task, but significant challenges remain—particularly detecting implicit toxicity and handling annotator subjectivity. The clear upward trend in our learning curves motivates scaling to the full dataset, and the gap between AUC and F1 motivates threshold optimization. These improvements, combined with generative data augmentation and adversarial testing, form our roadmap for the final submission.

References

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